

Dynamic Reinforcement Learning Control for Fractional-Order Neural Networks With Higher Order Interactions and Multiple Time Delays

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Abstract—Conventional neural networks’ reliance on pairwise connections limits their ability to model the modular higher order interactions inherent in biological neural systems. The stability analysis and control of large-scale networks that integrate these higher order interactions with fractional-order dynamics and multiple time delays present a significant challenge due to analytical intractability. This study establishes a theoretical and computational framework for a large-scale fractional-order neural network model featuring modular triplet clusters. Analytical criteria for stability and Hopf bifurcation are derived. To address the control challenge, a reinforcement learning (RL) framework is implemented for the adaptive tuning of higher order coupling strength to suppress destabilizing oscillations. Numerical results quantify the profound impact of system parameters on stability while demonstrating the RL controller’s ability to maintain

stability where traditional methods fail. By adaptively tuning the coupling strength without any prior knowledge of the system model, the proposed approach effectively suppresses oscillations, showcasing a advantage over model-based controllers for such high-dimensional, analytically intractable systems.

Impact Statement—Conventional neural networks are limited in modeling the complex modular interactions of biological systems, hindering their application in brain-inspired computing and large-scale neuromorphic engineering. Our work introduces a theoretical and control framework for fractional-order neural networks with higher order interactions and delays, which better mimics biological reality. By deriving stability criteria and employing a model-free RL controller to suppress oscillations, we enable robust and adaptive regulation of high-dimensional neural dynamics. This approach overcomes the analytical intractability of traditional methods, paving the way for more reliable and efficient neural network applications in areas such as autonomous systems, bio-inspired artificial intelligence, and the control of neuro-technological interfaces.

Index Terms—Fractional derivatives, higher order interactions, Hopf bifurcation, neural networks, reinforcement learning (RL).

Received 20 November 2025; revised 8 February 2026; accepted 9 March 2026. Date of publication 13 March 2026; date of current version 31 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62073172, and in part by the Natural Science Foundation of Jiangsu Province of China under Grant BK20221329. This article was recommended for publication by Associate Editor Qinglai Wei upon evaluation of the reviewers’ comments. (Corresponding author: Min Xiao.)

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Digital Object Identifier 10.1109/TAI.2026.3673682

I. INTRODUCTION

ARTIFICIAL neural networks (ANNs) are mathematical models constructed based on a profound understanding of the structure, composition, and operational mechanisms of biological neural networks, incorporating modern network theory and computational modeling methods [1], [2], [3]. Their primary objective is to emulate certain functional properties of the brain’s neural systems, such as parallel processing, self-organization, adaptability, robust learning, and associative memory capabilities. These capabilities have led to their widespread application across numerous domains. For example, ANNs have been used to: learn transfer models for accelerated optimization in dynamic environments [4]; design adaptive control laws for cyber-physical systems that can handle unknown nonlinearities [5]; and model gene regulatory networks as neural networks for robust control of biological processes [6]. Their successful deployment in solving such complex practical problems underscores the necessity for further investigation into the dynamical properties of ANNs.

Research on neural network dynamics has long focused on phenomena such as stability [7], bifurcation [8], and chaos [9], which are essential for understanding how neural systems process information and maintain functional integrity. Neuroscientific studies have linked abnormal neural rhythmicity to motor-control deficits, including Parkinson's disease. In ANNs, oscillatory activation patterns often emerge as a learned dynamical response when the network is repeatedly exposed to structured or periodic temporal inputs [10]. Neural dynamics typically manifest in two complementary regimes: stable attractor behavior that supports reliable memory retrieval, and oscillatory activity that facilitates temporal coding and sequential information processing. Time delays, arising from finite signal transmission speeds and synaptic processing, are inherent to both biological and ANNs [11], [12]. As information processing units, neural networks are generally expected to be stable and reliable; yet, delay-induced oscillatory dynamics serve both as a physiological basis for cognitive functions and as a source of system instability, performance degradation, and even pathological states. Among these dynamics, Hopf bifurcation represents a key mechanism governing the transition from stability to periodic oscillation [13], [14]. Therefore, understanding, predicting, and ultimately harnessing this stability-oscillation transition mechanism is of paramount importance for both theoretical and applied research.

The success of traditional ANNs in numerous fields largely stems from their powerful function approximation capabilities. However, their reliance on pairwise neuronal connections severely limits their ability to mimic the complex population coding and higher order information processing observed in biological neural systems. From a topological perspective, existing neuron interaction patterns are often organized into two canonical motifs: ring-like and star-like architectures [15], [16], [17], [18]. Ring configurations are frequently adopted due to their structural and dynamical significance in network design [15], [16], whereas star configurations rely on a hub node that enables efficient integration and coordination of distributed information flows [17], [18]. Biological evidence consistently identifies modular higher order interactions, in which neurons operate as coherent functional clusters, as a fundamental attribute of neural information processing [19], [20]. Incorporating higher order interactions allows neural network models to better reflect the hierarchical and modular properties of real-world systems. Meanwhile, fractional-order derivatives serve as a crucial mathematical tool for capturing history-dependent behaviors in neural dynamics due to their inherent memory and hereditary properties, enabling more precise modeling of key dynamics than integer order approaches [21], [22]. Recent advances have also addressed the stability and delay-handling control for discrete-time fractional-order delayed neural networks under uncertainties, further highlighting the active research on fractional-order delayed neural dynamics [23]. Constructing neural network models that integrate modular higher order interactions, fractional-order dynamics, and multiple time delays represents a critical pathway toward better approximating biological reality and enhancing computational performance.

Recent studies on neural-network-based modeling and control of nonlinear dynamical systems have also explored RNN-oriented identification and training strategies. Related learning-based modeling and control methodologies have also been investigated for nonlinear dynamical systems, including hybrid feed-forward recurrent architectures with Lyapunov-analyzed learning stability [24] and feedback-based hybrid optimization strategies to improve convergence quality and robustness in neural modeling [25]. These contributions highlight the effectiveness of neural learning in nonlinear system approximation and adaptive control. While the above studies demonstrate the strength of neural learning for identification and adaptive control, bifurcation-oriented stabilization in high-dimensional fractional-order delayed networks still faces the fundamental difficulty whose analytical threshold is often unavailable.

In high-dimensional complex systems, deriving explicit relationships between bifurcation thresholds and system parameters analytically is highly challenging and often infeasible. Traditional model-based control methods heavily rely on such closed-form expressions, significantly limiting their applicability in complex system scenarios. This is particularly true for neural networks incorporating fractional-order derivatives, higher order coupling, and multiple time delays, whose dynamical equations are highly nonlinear and high-dimensional, making it difficult for conventional control strategies to explicitly construct control laws for suppressing Hopf bifurcation and maintaining stability. To address this challenge, reinforcement learning (RL) has emerged as a data-driven intelligent control method with considerable advantages [26]. RL does not require an exact mathematical model of the system; instead, it enables an agent to learn optimal control policies autonomously through real-time interaction with the environment, thereby achieving adaptive regulation of complex dynamical systems. For instance, Li et al. [27] proposed a neural network-based approximation method for reinforcement signals to design adaptive optimal control for bounded nonlinear chaotic systems. Ding and Lei [28] employed deep RL to generate time-varying feedback gains, overcoming the odd-number limitation and model dependency of conventional delayed feedback control. Related learning-assisted control frameworks that couple rigorous controller synthesis with deep RL have also been reported for chaotic fractional-order systems with practical constraints [29]. In recent years, RL has achieved success in controlling various nonlinear dynamical systems. Its capability to handle high-dimensional state spaces, adapt to uncertainties, and cope with time-varying dynamics makes it particularly suitable for addressing bifurcation control problems in neural networks. Therefore, introducing RL into the bifurcation control of modular higher order fractional-order neural networks is not only theoretically feasible but also provides a novel technical pathway for the intelligent regulation of high-dimensional delayed dynamical systems. Currently, the study of RL for bifurcation control of dynamical systems remains largely unexplored.

To address these gaps, this study establishes an integrated theoretical and computational framework for analyzing and controlling large-scale fractional-order neural networks with

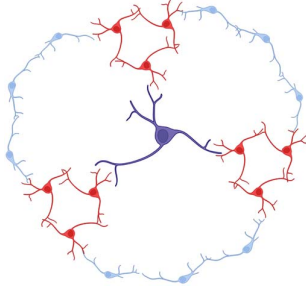


Fig. 1. Triple-neuron cluster neural network model.

modular higher order interactions and multiple time delays. Specifically, we aim to the following.

- 1) Propose a novel large-scale fractional-order neural network model that incorporates higher order coupling within a star-ring structure and modular triplet neuron clusters.
- 2) Conduct an in-depth stability and Hopf bifurcation analysis by employing Coates' flow-graph decomposition method [31] to handle the complex characteristic equation of the high-dimensional system. This analysis derives the critical stability conditions and bifurcation criteria with the time delay as the bifurcation parameter.
- 3) Numerically investigate the effects of key parameters (fractional order, self-feedback strength, higher order coupling strength, and network size) on the system's stability and bifurcation threshold.
- 4) Design and implement a RL-based control strategy. The key innovation of this controller is its online adaptive adjustment of the higher order coupling coefficient to intelligently suppress Hopf bifurcation, thereby demonstrating effective control without relying on explicit analytical expressions.

The remainder of this article is structured as follows. Section II details the mathematical model. Section III presents the stability and bifurcation analysis. Section IV introduces the RL controller design. Section V provides numerical simulations validating the theoretical results and demonstrating the efficacy of the RL controller. Finally, Section VI concludes the article and discusses key future research directions.

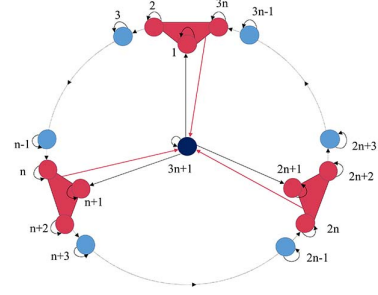
II. MATHEMATICAL MODEL

In this section, we construct a $3n + 1$ -dimensional delayed fractional neural network with a star-ring architecture and modular higher order coupling, as illustrated in Fig. 1. The network consists of $3n + 1$ neurons, including $3n$ peripheral neurons and one central neuron, and exhibits three symmetric groups of neuronal clusters. Each cluster comprises three neurons that convergently regulate the central neuron, as illustrated in Fig. 2. We adopt the Caputo derivative.

Definition 1 ([32]): The definition of the Caputo derivative is

$$D_t^\alpha x(t) = \frac{1}{\Gamma(\eta - \alpha)} \int_{t_0}^t (t - u)^{\eta - \alpha - 1} x^{(\eta)}(u) du$$

with $\alpha \geq 0$, $\alpha < \eta \leq \alpha + 1$ and $\eta \in \mathbb{Z}^+$.

Fig. 2. Schematic diagram of a $3n + 1$ -dimensional delayed neural network with a star-ring architecture and triplet neuron clusters.

Mathematically, this network can be described by the following set of higher order delay differential equations:

$$\begin{cases} D_t^\alpha x_1(t) = -\mu x_1(t) + a_{(3n)1} f(x_{3n}(t - \tau_{(3n)1})) \\ \quad + b_{(3n+1)1} f(x_{3n+1}(t - \tau_{(3n+1)1})) \\ D_t^\alpha x_{n+1}(t) = -\mu x_{n+1}(t) + a_{n(n+1)} f(x_n(t - \tau_{n(n+1)})) \\ \quad + b_{(3n+1)(n+1)} f(x_{3n+1}(t - \tau_{(3n+1)(n+1)})) \\ D_t^\alpha x_{2n+1}(t) = -\mu x_{2n+1}(t) \\ \quad + a_{2n(2n+1)} f(x_{2n}(t - \tau_{2n(2n+1)})) \\ \quad + b_{(3n+1)(2n+1)} f(x_{3n+1}(t - \tau_{(3n+1)(2n+1)})) \\ D_t^\alpha x_k(t) = -\mu x_k(t) + a_{(k-1)k} f(x_{k-1}(t - \tau_{(k-1)k})) \\ \quad k = 2, 3, \dots, n, n+2, \dots, 2n, 2n+2, \dots, 3n \\ D_t^\alpha x_{3n+1}(t) = -\mu x_{3n+1}(t) \\ \quad + c \left[g \left(\frac{x_{3n}(t - \delta) + x_1(t - \delta) + x_2(t - \delta)}{3} \right) \right. \\ \quad + g \left(\frac{x_n(t - \delta) + x_{n+1}(t - \delta) + x_{n+2}(t - \delta)}{3} \right) \\ \quad \left. + g \left(\frac{x_{2n}(t - \delta) + x_{2n+1}(t - \delta) + x_{2n+2}(t - \delta)}{3} \right) \right] \end{cases} \quad (1)$$

where $x_i(t)$ represents the state value of the i th neuron at time t , $\alpha \in (0, 1]$ denotes the fractional order, and μ is the self-feedback coefficient of the neurons. a_{ij} indicates the connection weight between neurons on the ring structure, while $b_{(3n+1)j}$ corresponds to the synaptic weight from the central neuron to the j th neuron. The time delay in information transmission from the i th to the j th neuron is denoted by $\tau_{ij} > 0$. The parameters c and $\delta > 0$ represent the higher order coupling strength and the higher order coupling delay, respectively. The activation functions $f(\cdot)$ and $g(\cdot)$ satisfy the following assumptions:

$$(H_1): f(0) = g(0) = 0, \quad f'(0) \neq 0, \quad g'(0) \neq 0.$$

Remark 1: In this study, the activation functions $f(\cdot)$ and $g(\cdot)$ are chosen as the hyperbolic tangent ($\tanh$) function, which satisfies the smoothness and boundedness assumptions required for the stability and bifurcation analysis. This choice is motivated by several theoretical and practical considerations: 1) the $\tanh$ function is smooth and exhibits a significant derivative near the origin, which facilitates the linearization of system dynamics around equilibria and simplifies the derivation of characteristic equations essential for the stability and Hopf bifurcation analysis conducted [7], [14]; 2) its saturating nature $\tanh(x) \in (-1, 1)$ aligns qualitatively with the firing rate saturation observed in biological neurons, making it a plausible

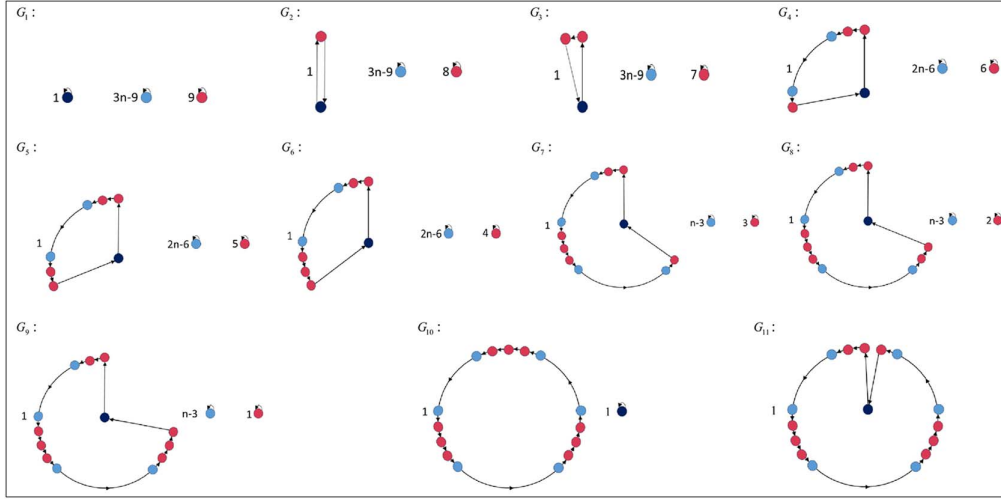


Fig. 3. Decomposition of flow graph.

candidate for modeling neural response in rate-based networks [16]; and 3) compared to the sigmoid function, the hyperbolic tangent exerts a stronger stretching effect on input data [30]. Nevertheless, we acknowledge that this choice represents a simplification of biological reality. Future work may explore adaptive or learnable activation functions to better capture biophysical realism while retaining analytical tractability where possible.

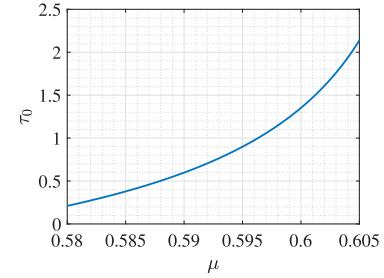
Clearly, the origin is an equilibrium point of system (1). The corresponding linearized system at the zero equilibrium is given by the following equation:

$$\begin{cases} D_t^\alpha x_1(t) = -\mu x_1(t) + p_{(3n)1} x_{3n}(t - \tau_{(3n)1}) \\ \quad + q_{(3n+1)1} x_{3n+1}(t - \tau_{(3n+1)1}) \\ D_t^\alpha x_{(n+1)}(t) = -\mu x_{(n+1)}(t) + p_{n(n+1)} x_n(t - \tau_{n(n+1)}) \\ \quad + q_{(3n+1)(n+1)} x_{3n+1}(t - \tau_{(3n+1)(n+1)}) \\ D_t^\alpha x_{(2n+1)}(t) = -\mu x_{(2n+1)}(t) \\ \quad + p_{2n(2n+1)} x_{2n}(t - \tau_{2n(2n+1)}) \\ \quad + q_{(3n+1)(2n+1)} x_{3n+1}(t - \tau_{(3n+1)(2n+1)}) \\ D_t^\alpha x_k(t) = -\mu x_k(t) + p_{(k-1)k} x_{k-1}(t - \tau_{(k-1)k}), \\ \quad k = 2, 3, \dots, n, n+2, \dots, 2n, 2n+2, \dots, 3n \\ D_t^\alpha x_{3n+1}(t) = -\mu x_{3n+1}(t) \\ \quad + h(x_{3n}(t - \delta) + x_1(t - \delta) + x_2(t - \delta) \\ \quad + x_n(t - \delta) + x_{n+1}(t - \delta) + x_{n+2}(t - \delta) \\ \quad + x_{2n}(t - \delta) + x_{2n+1}(t - \delta) \\ \quad + x_{2n+2}(t - \delta)) \end{cases} \quad (2)$$

where $p_{ij} = a_{ij} f'(0)$, $q_{ij} = b_{ij} f'(0)$, and $h = (1/3)cg'(0)$. Based on assumptions (H_1) , the characteristic matrix $\Delta(s)$ of system (2) can be computed.

To solve the characteristic equation of the high-dimensional system, a lemma derived from Coates' flow diagram is introduced.

Lemma 1 ([31]): Let G be a Coates' flow diagram corresponding to an n -dimensional square matrix L . The determinant

Fig. 4. Relationship curve between self-feedback coefficient μ and bifurcation threshold τ_0 .

of L can be expressed as follows:

$$\det L = \sum_{j=1}^k (-1)^{n+c_j} \Pi_j$$

where k denotes the total number of disjoint flow subgraphs in G , n is the order of the matrix L , and c_j indicates the number of disjoint directed cycles in the j th flow subgraph. The term Π_j represents the product of all edge weights within the j th flow subgraph.

This article makes the following assumption regarding the time delays:

$$\begin{aligned} (H_2) : & \tau_{(3n)1} = \tau_{i(i+1)} = \tau_1, \quad i = 1, 2, \dots, 3n-1 \\ & \tau_{(3n+1)1} = \tau_{(3n+1)(n+1)} = \tau_{(3n+1)(2n+1)} = \tau_2 \\ & \tau_1 = \tau_2 = \delta = \tau. \end{aligned}$$

Remark 2: For systems with heterogeneous delays, the characteristic equation will contain multiple distinct exponential terms, rendering the derivation of closed-form stability thresholds analytically intractable in general cases. Under the uniform delay assumption (H_2) , the Coates flow-graph method can be employed to obtain a tractable characteristic equation, thereby facilitating further investigation into stability criteria and Hopf bifurcation conditions. Consequently, this simplified approach

has been widely adopted by many researchers to study the stability and Hopf bifurcation of neural networks and to reveal their dynamical mechanisms [7], [13], [14]. Therefore, hypothesis (H_2) is acceptable. For heterogeneous systems with different delays, the mathematical framework is still valid conceptually, but the explicit expression of bifurcation point will depend on the specific delay distribution. Therefore, the exact formula in Section III cannot be directly extended to any delay configuration. However, they provide basic insights on how time delay, fractional order and coupling strength jointly affect stability.

We now derive the characteristic equation of system (2) under assumptions (H_1) and (H_2) by Lemma 1. The matrix $\Delta(s)$ corresponds to a Coates' flow diagram G . Based on topological similarity, all flow subgraphs of G can be classified into 11 distinct categories, as illustrated in Fig. 3. The coefficient M_i of the product of the connection gains for all edges corresponding to subgraph G_i ($i = 1, 2, \dots, 11$) can be listed as follows:

$$\begin{aligned} M_1 &= N_1(s^\alpha + \mu)^{3n+1}, \quad M_2 = N_2e^{-2s\tau}(s^\alpha + \mu)^{3n-1} \\ M_3 &= N_3e^{-3s\tau}(s^\alpha + \mu)^{3n-2} \\ M_4 &= N_4e^{-(n+1)s\tau}(s^\alpha + \mu)^{2n} \\ M_5 &= N_5e^{-(n+2)s\tau}(s^\alpha + \mu)^{2n-1} \\ M_6 &= N_6e^{-(n+3)s\tau}(s^\alpha + \mu)^{2n-2} \\ M_7 &= N_7e^{-(2n+1)s\tau}(s^\alpha + \mu)^n \\ M_8 &= N_8e^{-(2n+2)s\tau}(s^\alpha + \mu)^{n-1} \\ M_9 &= N_9e^{-(2n+3)s\tau}(s^\alpha + \mu)^{n-2} \\ M_{10} &= N_{10}(s^\alpha + \mu), \quad M_{11} = N_{11}e^{-(3n+1)s\tau} \end{aligned}$$

where

$$\begin{aligned} N_1 &= 1 \\ N_2 &= -h(q_{(3n+1)1} + q_{(3n+1)(n+1)} + q_{(3n+1)(2n+1)}) \\ N_3 &= -h(q_{(3n+1)1}p_{12} + q_{(3n+1)(n+1)}p_{(n+1)(n+2)} \\ &\quad + q_{(3n+1)(2n+1)}p_{(2n+1)(2n+2)}) \\ N_4 &= -h\left(q_{(3n+1)1}\prod_{i=1}^{n-1}p_{i(i+1)} + q_{(3n+1)(n+1)}\prod_{i=n+1}^{2n-1}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}\right) \\ N_5 &= -h\left(q_{(3n+1)1}\prod_{i=1}^np_{i(i+1)} + q_{(3n+1)(n+1)}\prod_{i=n+1}^{2n}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\right) \\ N_6 &= -h\left(q_{(3n+1)1}\prod_{i=1}^{n+1}p_{i(i+1)} + q_{(3n+1)(n+1)}\prod_{i=n+1}^{2n+1}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}p_{12}\right) \\ N_7 &= -h\left(q_{(3n+1)1}\prod_{i=1}^{2n-1}p_{i(i+1)} + q_{(3n+1)(n+1)}\prod_{i=n+1}^{3n-1}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\prod_{j=1}^{n-1}p_{j(j+1)}\right) \\ N_8 &= -h\left(q_{(3n+1)1}\prod_{i=1}^{2n}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(n+1)}\prod_{i=n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\prod_{j=1}^np_{j(j+1)}\right) \\ N_9 &= -h\left(q_{(3n+1)1}\prod_{i=1}^{2n+1}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(n+1)}\prod_{i=n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}p_{12}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\prod_{j=1}^{n+1}p_{j(j+1)}\right) \\ N_{10} &= (-1)^np_{(3n)1}\prod_{i=2}^{3n}p_{(i-1)i} \\ N_{11} &= -h\left(q_{(3n+1)1}\prod_{i=1}^{3n-1}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(n+1)}\prod_{i=n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\prod_{i=1}^{n-1}p_{i(i+1)}\right. \\ &\quad \left.+ q_{(3n+1)(2n+1)}\prod_{i=2n+1}^{3n-1}p_{i(i+1)}p_{(3n)1}\prod_{j=1}^{2n-1}p_{j(j+1)}\right). \end{aligned}$$

$$\Delta = \begin{bmatrix} s^\alpha + \mu & \cdots & 0 & \cdots & 0 & \cdots & -q_{(3n+1)1}e^{-s\tau(3n+1)1} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & s^\alpha + \mu & \cdots & 0 & \cdots & -q_{(3n+1)(n+1)}e^{-s\tau(3n+1)(n+1)} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & s^\alpha + \mu & \cdots & -q_{(3n+1)(2n+1)}e^{-s\tau(3n+1)(2n+1)} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ -he^{-s\delta} & \cdots & -he^{-s\delta} & \cdots & -he^{-s\delta} & \cdots & s^\alpha + \mu \end{bmatrix}_{(3n+1)^2}.$$

Hence, the characteristic equation of system (2) is given by the following equation:

$$\begin{aligned} & (s^\alpha + \mu)^{3n+1} + N_2(s^\alpha + \mu)^{3n-1}e^{-2s\tau} \\ & + N_3(s^\alpha + \mu)^{3n-2}e^{-3s\tau} + N_4(s^\alpha + \mu)^{2n}e^{-(n+1)s\tau} \\ & + N_5(s^\alpha + \mu)^{2n-1}e^{-(n+2)s\tau} + N_6(s^\alpha + \mu)^{2n-2}e^{-(n+3)s\tau} \\ & + N_7(s^\alpha + \mu)^ne^{-(2n+1)s\tau} + N_8(s^\alpha + \mu)^{n-1}e^{-(2n+2)s\tau} \\ & + N_9(s^\alpha + \mu)^{n-2}e^{-(2n+3)s\tau} + N_{10}(s^\alpha + \mu)e^{-3ns\tau} \\ & + N_{11}e^{-(3n+1)s\tau} = 0. \end{aligned} \quad (3)$$

III. LOCAL STABILITY AND HOPF BIFURCATION

This section analyzes the distribution of the roots of (3) and derives the conditions ensuring the stability of system (1) and the occurrence of a Hopf bifurcation.

By multiplying both sides of (3) by $e^{(3n+1)s\tau}$ and defining $\Psi(\tau) = (s^\alpha + \mu)e^{s\tau}$, we get the following:

$$\begin{aligned} & \Psi^{3n+1} + N_2\Psi^{3n-1} + N_3\Psi^{3n-2} + N_4\Psi^{2n} + N_5\Psi^{2n-1} \\ & + N_6\Psi^{2n-2} + N_7\Psi^n + N_8\Psi^{n-1} + N_9\Psi^{n-2} \\ & + N_{10}\Psi + N_{11} = 0. \end{aligned} \quad (4)$$

The general form of the roots of (4) is defined as follows:

$$\Psi_j = y_j + iz_j, \quad j = 1, 2, \dots, 3n+1 \quad (5)$$

where y_j and z_j represent the real and imaginary parts, respectively, of the j th root Ψ_j of (4). Clearly, each Ψ_j can be computed from the coefficients $N_0, N_1, \dots, N_{11}$.

A. Case Without Time Delay

When $\tau = 0$, $\Psi(0) = s^\alpha + \mu$, and the characteristic (4) transforms into

$$\begin{aligned} & \Psi(0)^{3n+1} + N_2\Psi(0)^{3n-1} + N_3\Psi(0)^{3n-2} + N_4\Psi(0)^{2n} \\ & + N_5\Psi(0)^{2n-1} + N_6\Psi(0)^{2n-2} + N_7\Psi(0)^n + N_8\Psi(0)^{n-1} \\ & + N_9\Psi(0)^{n-2} + N_{10}\Psi(0) + N_{11} = 0. \end{aligned} \quad (6)$$

Equation (6) is a polynomial equation of degree $3n+1$ in terms of $s^\alpha + \mu$, and thus possesses $3n+1$ complex roots, denoted as

$$s_j^\alpha + \mu = y_j + iz_j, \quad j = 1, 2, \dots, 3n+1.$$

The following assumption is made:

$$(H_3): \forall j \in \{1, 2, \dots, 3n+1\}, \mu > y_j.$$

Lemma 2: When $\tau = 0$, if assumptions (H_1) , (H_2) , and (H_3) are all satisfied, then system (1) is locally asymptotically stable at the origin.

Proof: When assumptions (H_1) , (H_2) , and (H_3) are all satisfied, it holds for every j that $\mu - y_j > 0$, which satisfies

$$\operatorname{Re}\{s_j^\alpha\} = y_j - \mu < 0.$$

This indicates that all characteristic roots of (3) have negative real parts, and thus system (1) is locally asymptotically stable at the origin. ■

B. Case With Time Delay

When $\tau > 0$, the delay τ is chosen as the bifurcation parameter to study the Hopf bifurcation of system (1). Assume that the characteristic (3) has a pair of pure imaginary roots $s = \pm i\omega$ ($\omega > 0$). Then $i\omega$ must satisfy

$$\begin{aligned} & \left[\omega^\alpha \left(\cos\left(\frac{\alpha\pi}{2}\right) + i \sin\left(\frac{\alpha\pi}{2}\right) \right) + \mu \right] (\cos(\omega\tau) + i \sin(\omega\tau)) \\ & = y_j + iz_j, \quad j = 1, 2, \dots, 3n+1. \end{aligned} \quad (7)$$

Separating the real and imaginary parts yields

$$\begin{cases} \left(\omega^\alpha \cos\left(\frac{\alpha\pi}{2}\right) + \mu \right) \cos(\omega\tau) - \omega^\alpha \sin\left(\frac{\alpha\pi}{2}\right) \sin(\omega\tau) = y_j \\ \left(\omega^\alpha \cos\left(\frac{\alpha\pi}{2}\right) + \mu \right) \sin(\omega\tau) + \omega^\alpha \sin\left(\frac{\alpha\pi}{2}\right) \cos(\omega\tau) = z_j \end{cases} \quad (8)$$

Rearranging gives

$$\begin{cases} \cos(\omega\tau) = \frac{y_j (\omega^\alpha \cos(\frac{\alpha\pi}{2}) + \mu) + z_j \omega^\alpha \sin(\frac{\alpha\pi}{2})}{(\omega^\alpha \cos(\frac{\alpha\pi}{2}) + \mu)^2 + (\omega^\alpha \sin(\frac{\alpha\pi}{2}))^2} := \Phi_1(\omega) \\ \sin(\omega\tau) = \frac{z_j (\omega^\alpha \cos(\frac{\alpha\pi}{2}) + \mu) - y_j \omega^\alpha \sin(\frac{\alpha\pi}{2})}{(\omega^\alpha \cos(\frac{\alpha\pi}{2}) + \mu)^2 + (\omega^\alpha \sin(\frac{\alpha\pi}{2}))^2} := \Phi_2(\omega). \end{cases} \quad (9)$$

Since $\cos^2(\omega\tau) + \sin^2(\omega\tau) = 1$, we obtain

$$\omega^{2\alpha} + 2\mu\omega^\alpha \cos\left(\frac{\alpha\pi}{2}\right) + \mu^2 - y_j^2 - z_j^2 = 0. \quad (10)$$

Let $\lambda = \omega^\alpha$ and define

$$\beta(\lambda) = \lambda^2 + 2\mu\lambda \cos\left(\frac{\alpha\pi}{2}\right) + \mu^2 - y_j^2 - z_j^2. \quad (11)$$

Clearly, if there exists a $\lambda > 0$ satisfying $\beta(\lambda) = 0$, then (10) possesses a corresponding positive root $\omega = \lambda^{(1/\alpha)} > 0$. The derivative is obtained as

$$\beta'(\lambda) = 2\lambda + 2\mu \cos\left(\frac{\alpha\pi}{2}\right). \quad (12)$$

Since $\mu > 0$ and $\alpha \in (0, 1]$, it follows that $\beta'(\lambda) > 0$ for all $\lambda > 0$, implying that $\beta'(\lambda)$ is monotonically increasing on the interval $\lambda \in (0, +\infty)$. We make the following assumptions:

$$(H_4): \forall j \in \{1, 2, \dots, 3n+1\}, \mu \geq \sqrt{y_j^2 + z_j^2}$$

$$(H_5): \exists j \in \{1, 2, \dots, 3n+1\}, \mu < \sqrt{y_j^2 + z_j^2}.$$

Lemma 3: Regarding (4), the following conclusion can be drawn.

- 1) If assumption (H_4) holds, then (4) has no purely imaginary roots.
- 2) If assumption (H_5) is satisfied, then (4) possesses at least one pair of purely imaginary roots.

Proof: The derivative $\beta'(\lambda)$ is monotonically increasing on the interval $\lambda \in (0, +\infty)$.

- 1) If assumption (H_4) holds, then $\beta(0) = \mu^2 - y_j^2 - z_j^2 \geq 0$. Hence, there exists no positive root λ satisfying $\beta(\lambda) = 0$, which implies that (10) has no positive solution $\omega = \lambda^{(1/\alpha)}$, and therefore (3) possesses no purely imaginary roots.

- 2) If assumption (H_5) holds, there exists some j such that $\beta(0) = \mu^2 - y_j^2 - z_j^2 < 0$. Moreover, since $\lim_{\lambda \rightarrow +\infty} \beta(\lambda) > 0$, it follows from the Intermediate Value Theorem that there exists at least one positive root λ such that $\beta(\lambda) = 0$. Consequently, (10) admits a positive root $\omega = \lambda^{(1/\alpha)}$, and (3) has a pair of purely imaginary roots $s = \pm i\omega$. ■

To maintain generality, suppose that (3) has a total of h pairs of purely imaginary roots, denoted as $\pm i\omega_i$ for $i = 1, 2, \dots, h$, where $h \geq 1$. From (9), we obtain

$$\tau_i^{(j)} = \begin{cases} \frac{\arccos \Phi_1(\omega_i) + 2j\pi}{\omega_i}, & \Phi_2(\omega_i) \geq 0 \\ \frac{2\pi - \arccos \Phi_1(\omega_i) + 2j\pi}{\omega_i}, & \Phi_2(\omega_i) < 0 \end{cases} \quad (13)$$

where $j = 0, 1, 2, \dots$. Let

$$k = \arg \min_i \tau_i^{(0)}, \quad i = 1, 2, \dots, h$$

then $\omega_0 = \omega_k$. The bifurcation threshold is defined as follows:

$$\tau_0 := \tau_k^{(0)}. \quad (14)$$

To verify the transversality condition, we make the following assumption:

$$(H_6): \psi_1\phi_1 + \psi_2\phi_2 > 0$$

where

$$\begin{aligned} \psi_1 &= -\alpha\omega_0^{(\alpha-2)} \cos\left(\frac{(\alpha-2)\pi}{2}\right), \phi_1 = \mu + \omega_0^\alpha \cos\left(\frac{\alpha\pi}{2}\right) \\ \psi_2 &= -\alpha\omega_0^{(\alpha-2)} \sin\left(\frac{(\alpha-2)\pi}{2}\right), \phi_2 = \omega_0^\alpha \sin\left(\frac{\alpha\pi}{2}\right). \end{aligned}$$

Lemma 4: Suppose $s(\tau) = u(\tau) + i\omega(\tau)$ is a root of (3) such that $u(\tau_0) = 0$ and $\omega(\tau_0) = \omega_0$. Under assumption (H_6) , the transversality condition holds as follows:

$$\operatorname{Re} \left[\frac{ds(\tau)}{d\tau} \right]_{s=i\omega_0, \tau=\tau_0} > 0.$$

Proof: Differentiating both sides of (4) with respect to τ yields

$$\frac{ds}{d\tau} = \frac{-\varepsilon(s)(s^\alpha + \mu)e^{s\tau}s}{\varepsilon(s)\alpha s^{\alpha-1}e^{s\tau} + \varepsilon(s)(s^\alpha + \mu)e^{s\tau}\tau}$$

where

$$\begin{aligned} \varepsilon(s) &= (3n+1)\Psi^{3n} + (3n-1)N_2\Psi^{3n-2} \\ &\quad + (3n-2)N_3\Psi^{3n-3} + 2nN_4\Psi^{2n-1} \\ &\quad + (2n-1)N_5\Psi^{2n-2} + (2n-2)N_6\Psi^{2n-3} \\ &\quad + nN_7\Psi^{n-1} + (n-1)N_8\Psi^{n-2} \\ &\quad + (n-2)N_9\Psi^{n-3} + N_{10}. \end{aligned}$$

It follows evidently that

$$\operatorname{sign} \left\{ \operatorname{Re} \left[\frac{ds(\tau)}{d\tau} \right]_{s=i\omega_0, \tau=\tau_0} \right\}$$

$$\begin{aligned} &= \operatorname{sign} \left\{ \operatorname{Re} \left[\frac{ds(\tau)}{d\tau} \right]_{s=i\omega_0, \tau=\tau_0}^{-1} \right\} \\ &= \operatorname{sign} \left\{ \operatorname{Re} \left[\frac{-\alpha s^{\alpha-1} - (s^\alpha + \mu)\tau}{(s^\alpha + \mu)s} \right]_{s=i\omega_0, \tau=\tau_0} \right\} \\ &= \operatorname{sign} \left\{ \frac{\psi_1\phi_1 + \psi_2\phi_2}{\phi_1^2 + \phi_2^2} \right\}. \end{aligned}$$

When assumption (H_6) holds, it follows $\operatorname{Re}[(ds(\tau)/d\tau)]_{s=i\omega_0, \tau=\tau_0} > 0$. ■

Lemma 5: If assumptions (H_1) and (H_2) hold, the following conclusions can be drawn for system (1).

- 1) If assumptions (H_3) and (H_4) are satisfied, then the zero equilibrium of system (1) is locally asymptotically stable for all delays $\tau \in [0, +\infty)$.
- 2) If assumptions (H_3) , (H_5) , and (H_6) all hold, then the zero equilibrium of system (1) is locally asymptotically stable when $\tau \in [0, \tau_0]$, and a Hopf bifurcation occurs when $\tau \in (\tau_0, +\infty)$.

IV. RL-BASED DECISION AND CONTROL FOR HIGHER ORDER COUPLING COEFFICIENTS

As established in the preceding analysis, the stability of system (1) is fundamentally determined by the distribution of the roots of its linearized characteristic (3). It is noteworthy that the coefficients of this characteristic equation explicitly incorporate the higher order coupling strength c . This mathematical relationship reveals a crucial control pathway: by dynamically adjusting the coupling coefficient c , one can directly influence the distribution of the characteristic roots, thereby actively steering the dynamic state of the system into a stable region or delaying the onset of bifurcation.

Remark 3: In high-dimensional networks with multiple time delays, fractional-order derivatives, and modular (higher order) interaction structures, the characteristic equation is typically transcendental and its roots depend on system parameters through highly coupled nonlinear terms. As a result, the explicit functional relationship becomes exceedingly complex and is generally intractable to derive in closed form, especially under real-time constraints. This poses a fundamental practical challenge for conventional robust/adaptive model-based controllers that require repeated online computation of characteristic roots or stability margins for parameter tuning [16]. By contrast, the proposed RL controller treats the plant as a black-box environment and learns an online adaptation law directly from interaction data, thereby avoiding explicit root-solving during deployment. This does not imply universal performance superiority over all model-based designs; rather, it highlights a deployment-time advantage in terms of computational burden and model-dependence for complex fractional-order delayed neural dynamics.

In this section, a data-driven RL framework is introduced. This approach enables an agent to learn an optimal control policy solely through real-time interaction with the environment, autonomously adjusting the higher order coupling coefficient c to suppress Hopf bifurcation and maintain system stability.

The subsequent sections detail the agent's design, training procedure, and control performance.

Remark 4: The control strategy of adaptively tuning the higher order coupling strength c is physically well-founded in both biological and artificial neural systems. In biological neural networks, the effective strength of such higher order interactions is not fixed. It can be dynamically modulated through mechanisms like activity-dependent synaptic plasticity and neuromodulation [33]. In parallel, within ANNs, higher order interactions can be explicitly defined and dynamically adjusted as structural parameters. For instance, hypergraph neural networks model hyperedges whose weights are directly trainable or externally tunable [34]. Therefore, modulating c corresponds to biologically plausible forms of slow, structural adaptation and is also technically feasible in neuromorphic or artificial network implementations. Compared to a traditional feedback gain, which injects a continuous external signal and requires high-precision real-time state measurement, adjusting c represents an internal, parametric intervention that alters the system's intrinsic interaction landscape. While traditional feedback acts rapidly but demands persistent energy input, tuning c can lead to a self-sustaining stable regime with lower ongoing cost, albeit on a slower timescale.

A. Basics of RL

RL is a machine learning methodology in which an agent learns optimal decision-making policies through interactions with an environment [28]. The core framework involves the agent observing the environmental state s_t at discrete time step t , and executing an action a_t accordingly. The environment transitions to a new state $s_{t+\Delta t}$ as a result of this action, while the agent receives a scalar reward $r_{t+\Delta t}$. This interactive process is typically formalized as a Markov decision process (MDP), with the objective of learning a policy π that maximizes the cumulative discounted reward $G_t = \sum_{k=0}^N \gamma^k r_{t+(k+1)\Delta t}$, where $\gamma \in [0, 1]$ is the discount factor, and N is the number of interactions. The interaction time T can be expressed as $T = N\Delta t$.

B. Temporal Difference (TD) Linear Agent Algorithm

Given the continuous nature of the control action (the higher order coupling coefficient c) and the dimensionality of the state space, we design a TD learning agent based on linear function approximation and polynomial features. This agent approximates the action-value function $Q(s, a)$ by constructing polynomial features from the state and action. Its policy employs an ϵ -greedy strategy to balance exploration and exploitation. Although the controlled neural network model is complex, its dynamics near the equilibrium point can be well characterized through linearization [35], [36].

Remark 5: Although the future state evolution of the fractional-order delayed neural network studied in this article depends on its historical states (due to the delay term τ), this does not contradict the use of the MDP framework to describe and control it. The fundamental reason is that the 'state' s_t in

an MDP is an abstract information set designed for decision-making. In this study, the environment (i.e., the neural network dynamic model) internally and autonomously handles all delay-dependent relationships through its numerical integration process after receiving the agent's action a_t (the adjustment of the higher order coupling coefficient). It then computes the next system state $s_{t+\Delta t}$ and the reward $r_{t+\Delta t}$. From the agent's perspective, the tuple $(s_t, a_t, r_{t+\Delta t}, s_{t+\Delta t})$ returned by the environment inherently satisfies the Markov property required by the MDP framework that the next state $s_{t+\Delta t}$ depends only on the current state s_t and the current action a_t . In other words, the internal dynamics of the environment (with delays) are treated as a "opaque box" that encapsulates all complex historical dependencies and exposes only a Markovian interface to the agent. Consequently, the agent can learn the optimal control policy by interacting with this environment without needing to explicitly observe or memorize historical states.

C. Controller Design

1) *State and Action:* The observed state s of the agent consists of a $(3n + 1)$ -dimensional neuronal state vector, i.e., $s = [x_1, x_2, \dots, x_{3n+1}]^T$. The control action a is the higher order coupling coefficient c applied directly to the system. Its value is constrained within $c \in [c_{\min}, c_{\max}]$ to prevent system instability and avoid control saturation.

2) *Feature Function:* To achieve effective approximation of the action-value function $Q(s, a)$, this article employs a linear function approximator based on polynomial feature construction. The feature function $\phi(s_t, a_t)$ maps the current state vector $s_t \in \mathbb{R}^{3n+1}$ and the action $a_t \in \mathbb{R}$ (i.e., the higher order coupling coefficient c) into a high-dimensional feature space. In this case, the approximated action-value function is $\hat{Q}(s, a, \theta) = \phi^T(s, a) \cdot \theta$, θ is the corresponding weight coefficient.

First, the state and action are concatenated into a joint vector

$$\mathbf{v} = [s_t; a_t] \in \mathbb{R}^{3n+2}.$$

Then, by incorporating a bias term, linear terms, quadratic terms, and cross terms, the polynomial feature vector is constructed as follows:

$$\phi(s_t, a_t) = [1, \mathbf{v}, \mathbf{v}^2, \{v_i v_j \mid 1 \leq i < j \leq 3n + 2\}]^T$$

where, 1 denotes the bias term; $\mathbf{v}$ represents the linear terms; $\mathbf{v}^2$ corresponds to the quadratic terms; $v_i v_j$ includes all pairwise cross terms.

3) *Reward Function:* The reward function designed in this article consists of two components: a tracking error reward r_{te} and an action penalty r_{ap}

$$r = r_{te} + r_{ap}.$$

The tracking reward r_{te} drives the system states x_i toward the target equilibrium point using a weighted Euclidean distance metric

$$r_{te} = -\rho_{te} \cdot \sqrt{\sum_{i=1}^{3n+1} \lambda_i (x_i - x_i^{\text{target}})^2}$$

where

$$\lambda_i = \begin{cases} 1.0, & i = 1, n+1, 2n+1, 3n+1 \\ 0.8, & \text{others.} \end{cases}$$

The λ_i represents state-dependent weighting coefficients that prioritize different neuronal states.

The action penalty r_{ap} adopts a U-shaped curve, which does not encourage excessive control actions, but instead encourages the smallest possible action values

$$r_{ap} = -\rho_{ap} \cdot \tanh(c)^2.$$

The symmetric penalty r_{ap} regularizes the control magnitude. The optimal sign for c is discovered through the tracking reward r_{te} , which strongly penalizes state deviations.

Remark 6: The control objective is not to model arbitrary global dynamics but to stabilize the system near the equilibrium point. In this local region, the system's nonlinear dynamics are dominated by its linearized model [13], [14], [16]. Therefore, the agent does not need to learn a globally complex value function. The linear function approximator with quadratic polynomial features is designed to capture the key features for stabilization near equilibrium. Linear terms $\mathbf{v}$ capture the first-order influence of states and action on value. Quadratic and cross terms $\mathbf{v}^2, v_i v_j$ directly correspond to the quadratic form common in Lyapunov functions for stability and captures the coupling effects between different neuronal states and between states and the control action c . These effects are central to influencing the bifurcation threshold. Thus, we are not using a simple model to blindly fit a complex function, but rather a structurally correct model to approximate the subspace of the value function that truly matters for stability control near equilibrium.

4) *Agent Architecture and Training:* The agent utilizes a linear approximator, whose weight vector ω is updated with $\rho_{te} = 10$, $\rho_{ap} = 1.8$, learning rate $\alpha_r = 0.1$, exploration rate $\epsilon = 0.1$, fractional order $\alpha = 1$, and discount factor $\gamma = 0.95$. The agent is trained by minimizing the TD error, with the ultimate goal of deriving the optimal control policy π^* through learning the optimal value function.

Remark 7: The parameter vector of the TD-linear Q-function approximator is initialized as $\theta_0 = \mathbf{0}$. This neutral initialization avoids imposing any preference on the sign or magnitude of the continuous action c ; the policy is shaped subsequently through ϵ -greedy exploration and TD updates in Algorithm 1. The subsequent RL control algorithm is presented in Algorithm 2.

V. EXAMPLES AND SIMULATIONS

In this section, through systematic numerical investigations, we demonstrate how key parameters influence system dynamics and further introduce a RL strategy for adaptive control of higher order coupling coefficients.

This section analyzes the effects of the fractional order α , the self-feedback coefficient μ , and the higher order coupling strength c on the stability of the equilibrium point and the bifurcation threshold τ_0 of system (1). Simulations are performed using the Caputo derivative definition, with the activation functions chosen as $f(x) = g(x) = \tanh(x)$. The time delay parameters are configured to satisfy assumption (H_2). A neural

Algorithm 1: Reinforcement Learning-based Adaptive Control for Hopf Bifurcation Suppression.

Require: Neural network model $\mathcal{M}$ with state dimension d_s , action dimension d_a ; Learning rate α_r , exploration rate ϵ , discount factor γ ; Action range $[c_{\min}, c_{\max}]$; Time delay τ , simulation parameters μ, p_{ij}, q_{ij} .

Ensure: Trained policy π^* for adaptive higher-order coupling coefficient control; Stabilized system states $x_1, \dots, x_{3n+1}$.

Initialize TDLearnerAgent with $\alpha_r, \epsilon, \gamma, [c_{\min}, c_{\max}]$

for episode = 1 to N_{episodes} **do**

 Initialize system states $x_1(0), \dots, x_{3n+1}(0)$ with predefined initial conditions

 Initialize history buffers for delayed states

 Set initial control action $c(0) = 0$

for $t = 0$ to T_{steps} **do**

 Observe current state $s_t = [x_1(t), \dots, x_{3n+1}(t)]$

 Select action a_t using ϵ -greedy policy:

 With probability ϵ : $a_t \leftarrow$

 random uniform sample from $[c_{\min}, c_{\max}]$

 Otherwise: $a_t \leftarrow \arg \max_a \hat{Q}(s_t, a, \theta_t)$ via candidate sampling

 Set control action: $c(t) \leftarrow a_t$

 Retrieve delayed states $s_{t-\tau}$ from history buffers

 Compute next state $s_{t+\Delta t}$:

$s_{t+\Delta t} \leftarrow \text{EnvironmentStep}(s_t, s_{t-\tau}, a_t, \mathcal{M}, \Delta t)$

 Compute reward r_t :

$r_t \leftarrow -\rho_{te} \cdot \sqrt{\sum_{i=1}^{3n+1} \lambda_i (x_i(t + \Delta t) - x_i^{\text{target}})^2} - \rho_{ap} \cdot \tanh(a_t)^2$

 Observe next state $s_{t+\Delta t}$

 Compute TD error and update policy weights:

$\delta_t \leftarrow r_t + \gamma \cdot \max_{a'} \hat{Q}(s_{t+\Delta t}, a', \theta_t) - \hat{Q}(s_t, a_t, \theta_t)$

$\theta_{t+\Delta t} \leftarrow \theta_t + \alpha_r \cdot \delta_t \cdot \phi(s_t, a_t)$

 Update history buffers with new state $s_{t+\Delta t}$

end for

end for

return Trained policy π^* parameterized by optimized weights θ^*

network of dimension 10 (i.e., $n = 3$) is considered, with fixed parameters as follows: $a_{12} = a_{45} = a_{56} = a_{78} = 0.5$, $b_{(10)1} = b_{(10)4} = b_{(10)7} = 0.1$, $a_{23} = a_{89} = 1.5$, $a_{67} = a_{91} = 0.3$, $a_{34} = -1$, $\alpha = 0.98$, $c = 0$. These values are selected to satisfy the modeling assumptions used in the stability/bifurcation analysis.

By computing the roots of (4), we get

$$\max_n \{\text{Re}(\Psi_n)\} = 0.5783, \quad \max_n \{|\Psi_n|\} = 0.6154.$$

Fig. 4 presents the response curve of the bifurcation threshold τ_0 with respect to the self-feedback coefficient μ .

A. Influence of the Self-Feedback Coefficient μ on System (1)

When $\mu = 0.63 > \max_n \{|\Psi_n|\} = 0.6154$, assumptions (H_1) – (H_4) are satisfied. Under these conditions, (3) has no

Algorithm 2: Deployment of Trained RL Agent for Hopf Bifurcation Suppression.

Require: Trained policy π^* parameterized by θ^* (via $\hat{Q}$ -function); Neural network model $\mathcal{M}$ with state dimension d_s , action dimension d_a ; Action range $[c_{\min}, c_{\max}]$; Time delay configuration τ (can be uniform or heterogeneous $\{\tau_i\}$); Simulation parameters μ, p_{ij}, q_{ij} ; Activation time $t_{\text{activation}}$; Total simulation time T_{final} .

Ensure: Controlled system states $x_1(t), \dots, x_{3n+1}(t)$ for $t \in [0, T_{\text{final}}]$; Adaptive control signal $c(t)$ for $t \geq t_{\text{activation}}$.

Initialize system states $x_1(0), \dots, x_{3n+1}(0)$ with predefined initial conditions

Initialize higher-order coupling coefficient $c(0) = 0$

Initialize history buffers for delayed states

▷ Phase 1: Without control

for $t = 0$ to $t_{\text{activation}}$ with step Δt **do**

Simulate system with fixed $c(t)$:

$s_{t+\Delta t} \leftarrow \text{EnvironmentStep}(s_t, s_{t-\tau}, c(t), \mathcal{M}, \Delta t)$

Update history buffers with $s_{t+\Delta t}$

$t \leftarrow t + \Delta t$

end for

▷ Phase 2: Aagent activated

for $t = t_{\text{activation}}$ to T_{final} with step Δt **do**

Observe current state $s_t = [x_1(t), \dots, x_{3n+1}(t)]$

Query trained policy to obtain control action:

$a_t \leftarrow \arg \max_{a \in [c_{\min}, c_{\max}]} \hat{Q}(s_t, a, \theta^*)$ ▷ Greedy action

selection

Apply control action with saturation:

$c(t) \leftarrow \text{clip}(a_t, c_{\min}, c_{\max})$

Simulate system with adaptive $c(t)$:

$s_{t+\Delta t} \leftarrow \text{EnvironmentStep}(s_t, s_{t-\tau}, c(t), \mathcal{M}, \Delta t)$

Update history buffers with $s_{t+\Delta t}$

$t \leftarrow t + \Delta t$

end for

Generate control performance plots and metrics

return State trajectories $\{x_i(t)\}$ and control signal $\{c(t)\}$

purely imaginary roots, and system (1) remains stable for any time delay $\tau \geq 0$. With $\tau = 0.5$ selected for simulation, Fig. 5 shows that all state variables $x_i(t)$ ($i = 1, 2, \dots, 10$) of system (1) converge to the zero equilibrium, which is consistent with Lemma 5.

When $\mu = 0.58 < \max_n \{|\Psi_n|\} = 0.6154$, assumptions $(H_1) - (H_3)$, (H_5) , and (H_6) are satisfied. Under this parameter setting, (3) possesses a pair of purely imaginary roots, and the system has a bifurcation threshold $\tau_0 = 0.2107$. Two delay values are considered: $\tau = 0.1 < \tau_0$ and $\tau = 0.3 > \tau_0$. As illustrated in Fig. 6, for $\tau = 0.1$, all system states converge rapidly to the zero equilibrium, confirming the stability of system (1). In contrast, Fig. 7 demonstrates that for $\tau = 0.3$, the system exhibits sustained oscillations, indicating the occurrence of a Hopf bifurcation and the emergence of periodic solutions.

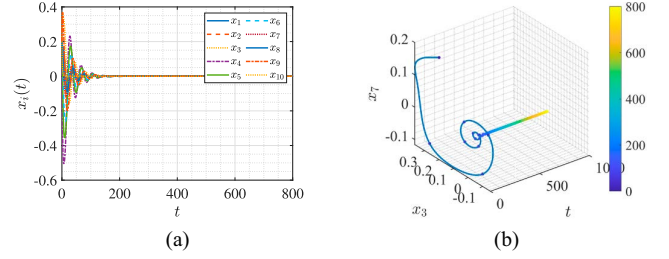


Fig. 5. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.63$ and $c = 0$. Asymptotic stability is observed at a time delay of $\tau = 0.50$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 .

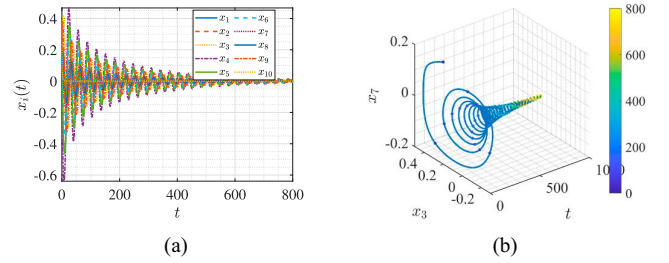


Fig. 6. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.58$ and $c = 0$. Asymptotic stability is observed at a time delay of $\tau = 0.10 < \tau_0 = 0.2107$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 .

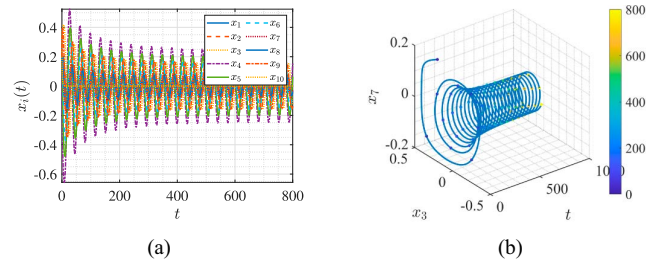


Fig. 7. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.58$ and $c = 0$. Periodic oscillation is observed at a time delay of $\tau = 0.30 > \tau_0 = 0.2107$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 .

B. Influences of the Higher Order Coupling Strength c and the Fractional Order α on the Bifurcation Threshold τ_0

With $\mu = 0.58$, the influences of the higher order coupling strength c and the fractional order α on the bifurcation threshold τ_0 are investigated.

Fig. 8 and Table I present the response surface and the corresponding relationships of the bifurcation threshold τ_0 with respect to the fractional order α and the higher order coupling coefficient c . Higher order interactions increase the delay stability margin of the system, leading to a larger Hopf bifurcation threshold and thereby enhancing system stability. When $c = -0.1$, the bifurcation threshold τ_0 increases from 0.2107 (at $c = 0$) to 0.5932. For the same time delay $\tau = 0.3$, the system

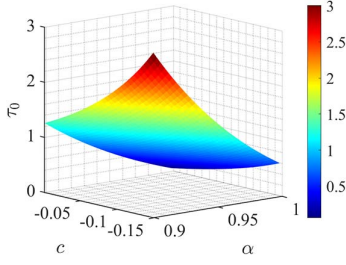


Fig. 8. Response surface of the bifurcation threshold τ_0 with respect to the fractional order α and the higher order coupling strength c at $\mu = 0.58$.

TABLE I
RELATIONSHIP BETWEEN HIGHER ORDER COUPLING COEFFICIENT c , FRACTIONAL ORDER α , AND τ_0 , ω_0

Coupling c	Fractional Order α	τ_0	ω_0
$c = 0$	0.91	1.0671	0.1149
	0.92	0.9139	0.1230
	0.93	0.7725	0.1316
	0.94	0.6418	0.1406
	0.95	0.5211	0.1502
	0.96	0.4095	0.1603
	0.97	0.3062	0.1708
	0.98	0.2107	0.1819
	0.99	0.1223	0.1935
	1.00	0.0404	0.2057
$c = -0.1$	0.91	1.8939	0.0901
	0.92	1.6565	0.0973
	0.93	1.4389	0.1051
	0.94	1.2392	0.1134
	0.95	1.0561	0.1223
	0.96	0.8882	0.1317
	0.97	0.7343	0.1418
	0.98	0.5932	0.1524
	0.99	0.4639	0.1637
	1.00	0.3453	0.1755

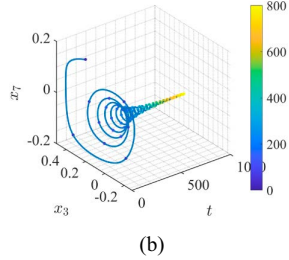
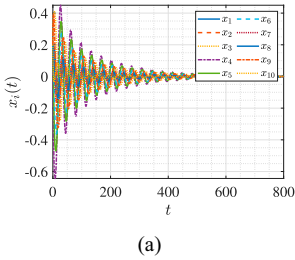


Fig. 9. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.58$ and $c = -0.1$. Asymptotic stability is observed at a time delay of $\tau = 0.30 < \tau_0 = 0.5932$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 .

is unstable when $c = 0$ (as shown in Fig. 7), whereas it remains stable when $c = -0.1$ (see Fig. 9). The systematic variation of τ_0 with fractional order α further emphasizes the flexibility offered by fractional calculus in tuning dynamic responses.

We fix all other parameters and set the delay to the bifurcation threshold at $c = 0$, $\alpha = 0.98$, $\tau = \tau_0 = 0.2107$ (see Table I), where the characteristic equation admits a purely imaginary pair $s = \pm i\omega_0$ with $\omega_0 \approx 0.1819$. Then we vary c in a neighborhood

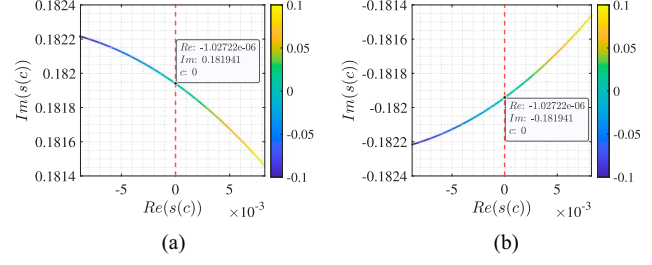


Fig. 10. Influence of the higher order coupling coefficient c on the distribution of characteristic roots. (a) Root locus of the complex conjugate root with positive imaginary part; at $c = 0$, $c \neq 0$, this root is purely imaginary. (b) Root locus of the complex conjugate root with negative imaginary part; at $c = 0$, $c \neq 0$, this root is purely imaginary. When $c = 0$, $c \neq 0$, these two roots form a complex conjugate pair.

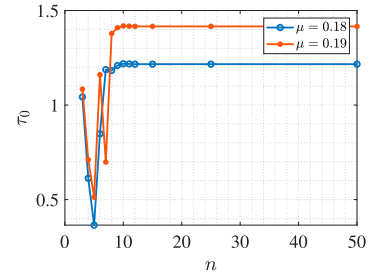


Fig. 11. Response curve of the bifurcation threshold τ_0 with respect to the neuron population size n at $\alpha = 0.95$ and $c = -0.5$.

of 0 and numerically compute the corresponding roots. The Fig. 10 (Re–Im plot with colorbar indicating c) shows that: at $c = 0$, the critical roots lie on the imaginary axis; for $c < 0$, the real parts of this conjugate pair become negative, and the roots move into the left half-plane; for $c > 0$, the real parts become positive, and the roots move into the right half-plane.

This provides direct numerical evidence that, near the Hopf point, decreasing c shifts the root locus across the imaginary axis in the stabilizing direction. This observation is also consistent with our bifurcation-threshold mapping: decreasing c enlarges the delay stability margin. Hence, for a fixed physical delay τ , the controller can stabilize the system by adapting c so that $\tau < \tau_0(c)$.

C. Influence of the Neuron Population Size n on the Bifurcation Threshold τ_0

With $\alpha = 0.95$, $c = -0.5$, the influence of the neuron population size n on the bifurcation threshold τ_0 is investigated. Fig. 11 and Table II display the response curves and corresponding relationships of the bifurcation threshold τ_0 with respect to the neuron population size n and the self-feedback coefficient μ .

The systematic variation of τ_0 with both n and μ indicates that larger networks may require different stability criteria compared to their smaller counterparts. This finding has implications for understanding biological neural systems, which typically involve large numbers of interconnected neurons.

TABLE II
RELATIONSHIP BETWEEN NEURON SCALE n ,
SELF FEEDBACK COEFFICIENT μ , AND τ_0 , ω_0

n	$\mu = 0.18$		$\mu = 0.19$	
	τ_0	ω_0	τ_0	ω_0
3	1.0426	0.5360	1.0838	0.5319
4	0.6128	0.3269	0.7109	0.3210
5	0.3650	0.2589	0.5127	0.2518
6	0.8482	0.1853	1.1601	0.1761
7	1.1872	0.2450	0.6983	0.0739
8	1.1826	0.2414	1.3781	0.2339
9	1.2092	0.2394	1.4088	0.2319
10	1.2181	0.2393	1.4182	0.2318
11	1.2173	0.2394	1.4170	0.2319
12	1.2162	0.2395	1.4159	0.2320
15	1.2162	0.2395	1.4159	0.2320
25	1.2162	0.2395	1.4159	0.2320
50	1.2162	0.2395	1.4159	0.2320

D. Higher Order Coupling Coefficient c in Decision-Making Control of RL

In high-dimensional complex systems, analytically deriving an explicit relationship between the higher order coupling coefficient c and the system's bifurcation threshold τ_0 is highly challenging and often infeasible. Traditional model-based control methods heavily rely on such closed-form expressions, significantly limiting their applicability in complex system scenarios. The data-driven control framework proposed in this article, based on RL, completely eliminates the dependence on explicit analytical models. Through real-time interaction between the agent and the environment, this framework autonomously learns and dynamically optimizes the higher order coupling coefficient c , thereby enabling intelligent suppression of Hopf bifurcation and stability regulation of the system.

Numerical experiments, conducted under two critical delay conditions ($\tau = 0.2$ and $\tau = 0.3$), provide evidence for the efficacy of this approach. The learning trajectory of the agent, as depicted in the total reward and TD-error curves (Figs. 12 and 13). The initial high TD-error and volatile rewards signify a phase of active exploration, where the agent tests various actions to understand the environment's response. Subsequently, the monotonic increase in cumulative reward and the decay of TD-error indicate a successful transition to exploitation, where the agent stabilizes around a policy that effectively balances state regulation with control effort. Demonstrates a clear convergence toward an optimal policy.

When $n = 3$, the dimension of this feature function is

$$\dim(\phi) = 1 + 11 + 11 + C_{11}^2 = 78.$$

The heatmaps of weight updates (Fig. 14) offer a granular view of the agent's learning process within the high-dimensional state-action space. Notably, the distinct weight configurations for $\tau = 0.2$ and $\tau = 0.3$ underscore the agent's capacity to discern and adapt to the unique dynamical regimes induced by different delays.

The validation of the RL framework is illustrated in Figs. 15 and 16. Prior to the agent's intervention ($t < 800$), the system with its inherent higher order coupling strength ($c = 0$), exhibits persistent oscillations, confirming the analytical prediction of

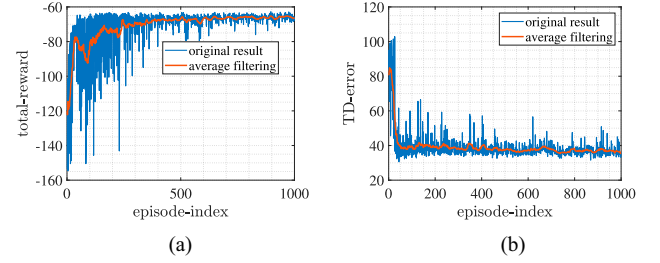


Fig. 12. Learning curve of RL controller when $\tau = 0.2$, $\alpha = 1$ and $\mu = 0.58$. Blue and red traces correspond to raw episode returns and their smoothed counterpart via moving-average filtering, respectively. (a) Reward curve of the agent. (b) TD-error curve of the agent.

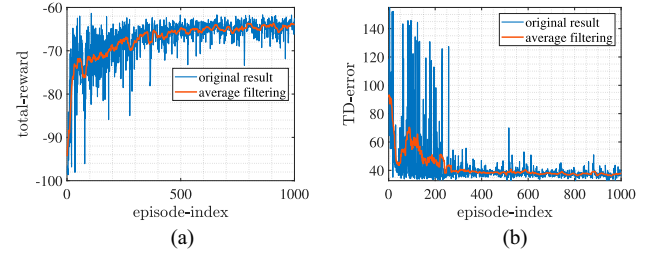


Fig. 13. Learning curve of RL controller when $\tau = 0.3$, $\alpha = 1$ and $\mu = 0.58$. Blue and red traces correspond to raw episode returns and their smoothed counterpart via moving-average filtering, respectively. (a) Reward curve of the agent. (b) TD-error curve of the agent.

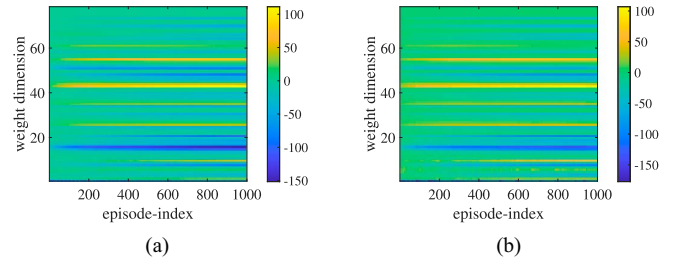


Fig. 14. Heatmap of weight updates in the agent's state-action space. (a) Scenario with the parameter $\tau = 0.2$. (b) Scenario with the parameter $\tau = 0.3$.

a Hopf bifurcation at the given delays. The activation of the RL controller at $t = 800$ triggers an immediate and intelligent adjustment of c (Figs. 17 and 18). The agent does not simply apply a maximal or minimal control signal; instead, it learns a nuanced policy, dynamically tuning c to a specific value that rapidly quenches the oscillations and drives the system states to the equilibrium. This successful suppression of bifurcation, achieved without any prior knowledge of the system's transcendental characteristic equation, highlights the advantage of RL in dealing with analytically intractable high-dimensional systems.

To verify the generalization capability of the proposed RL controller in more biologically plausible scenarios, we design zero-shot generalization experiments to test its performance in environments with heterogeneous delays. The delay τ_{ij} of each neuron x_i is defined as τ_i , where j is the index of each neuron connected to x_i . We assign each neuron a different delay value

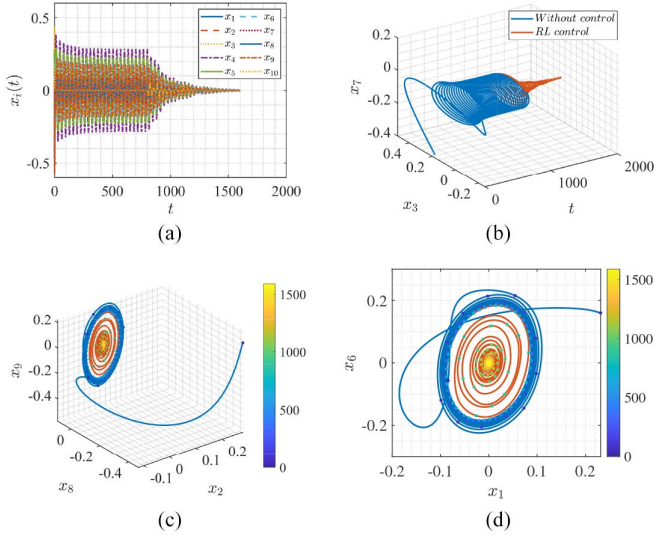


Fig. 15. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.58$ and $\tau = 0.2$. Hopf bifurcation suppression via intelligent agent-controlled higher order coupling strength: $c = 0$ for $t < 800$ and RL-based regulation of c after $t = 800$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 . (c) 3-D phase portrait in the (x_2, x_8, x_9) subspace. (d) 2-D phase diagram in the (x_1, x_6) plane.

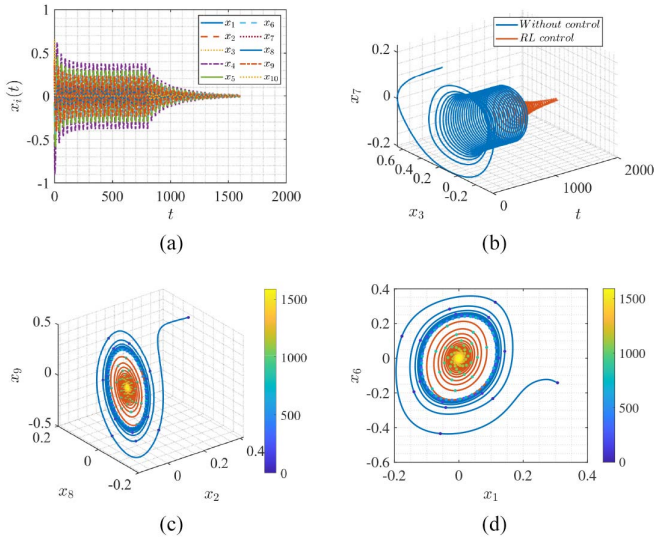


Fig. 16. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.58$ and $\tau = 0.3$. Hopf bifurcation suppression via intelligent agent-controlled higher order coupling strength: $c = 0$ for $t < 800$ and RL-based regulation of c after $t = 800$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 . (c) 3-D phase portrait in the (x_2, x_8, x_9) subspace. (d) 2-D phase diagram in the (x_1, x_6) plane.

randomly within $\pm 30\%$ of the training delay τ , constructing a test environment that violates the uniform delay assumption (H_2). It is noteworthy that the controller is deployed without any retraining in this environment, directly using the strategy trained under uniform delay.

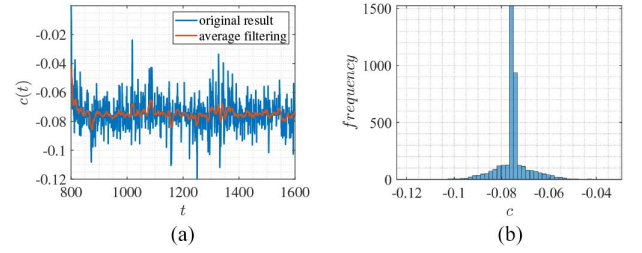


Fig. 17. Adaptive regulation and action distribution of the higher order coupling coefficient c by the RL controller when $\tau = 0.2$ for $t > 800$. (a) Temporal evolution of the coupling coefficient c , showcasing the agent's intervention starting at $t = 800$ to suppress system oscillations. (b) Distribution frequency of the control action c , illustrating the preference of the learned policy for specific coupling strength values.

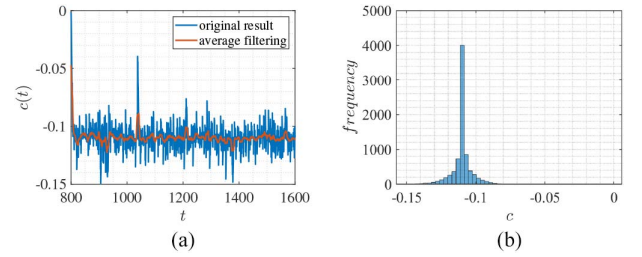


Fig. 18. Adaptive regulation and action distribution of the higher order coupling coefficient c by the RL controller when $\tau = 0.3$ for $t > 800$. (a) Temporal evolution of the coupling coefficient c , showcasing the agent's intervention starting at $t = 800$ to suppress system oscillations. (b) Distribution frequency of the control action c , illustrating the preference of the learned policy for specific coupling strength values.

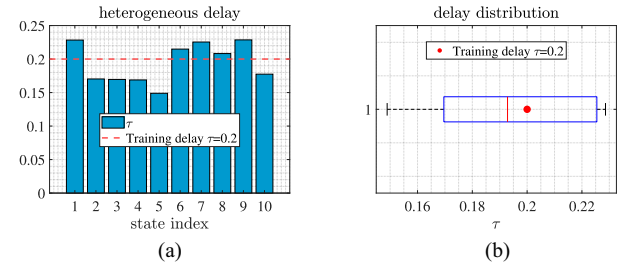


Fig. 19. Heterogeneous delay configuration for training delay $\tau = 0.2$. (a) Bar chart of each neuron's delay value τ_i , with the red dashed line indicating the training delay $\tau = 0.2$. (b) Box plot of the delay distribution.

Fig. 19 shows the heterogeneous delay configuration for the training delay $\tau = 0.2$, where each bar represents the delay value τ_i of a neuron, and the red dashed line indicates the training delay $\tau = 0.2$. It can be seen that the delays of each neuron are randomly distributed between 0.15 and 0.23, covering a range of $\pm 30\%$, simulating the natural diversity of synaptic delays in biological neural networks. Correspondingly, Fig. 20 shows a similar configuration for the training delay $\tau = 0.3$, with delay values varying between 0.24 and 0.37. Both configurations break the uniform delay assumption relied upon in the theoretical analysis, providing a test for evaluating the robustness of the controller.

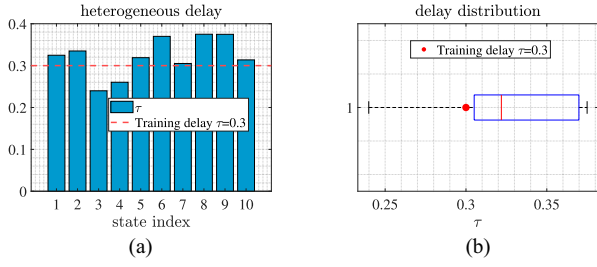


Fig. 20. Heterogeneous delay configuration for training delay $\tau = 0.3$. (a) Bar chart of each neuron's delay value τ_i , with the red dashed line indicating the training delay $\tau = 0.3$. (b) Box plot of the delay distribution.

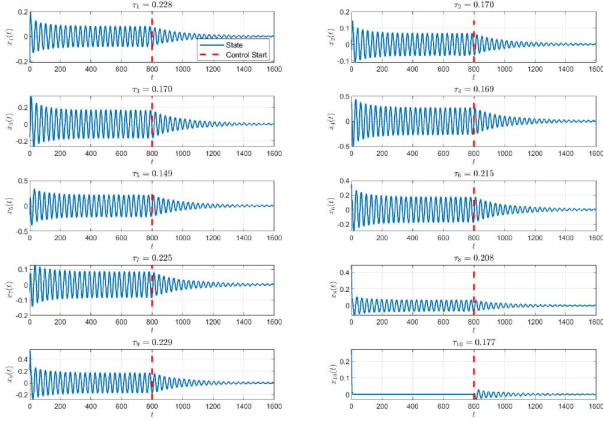


Fig. 21. Performance of the controller trained with $\tau = 0.2$ in a heterogeneous delay environment.

As shown in Figs. 21 and 22, before the control is activated, the states of each neuron exhibit sustained oscillations, indicating that the heterogeneous delay does not alter the essential characteristic of the system undergoing Hopf bifurcation at supercritical delays, but the oscillation patterns become more complex due to the dispersion of delay distribution. Notably, when the controller is activated, all state variables are rapidly stabilized to the equilibrium point, and oscillations are effectively suppressed. This result proves that despite the significant difference in delay structure between the training and testing environments, the stabilization strategy learned by the RL controller has generalization ability and can adapt to the heterogeneity of delays.

Figs. 23(a) and 24(a) show the change of control parameter c over time for $\tau = 0.2 * (1 \pm 30\%)$ and $\tau = 0.3 * (1 \pm 30\%)$. It is noteworthy that although the delay configurations are different, the final steady-state values of c in both cases fall in the negative range, indicating that the controller learns a general principle of regulating the higher order coupling strength through negative feedback, rather than an overfitting strategy for a specific delay configuration. The histograms in Figs. 23(b) and 24(b) further quantify the distribution of the control parameter.

Remark 8: Although the theoretical analysis is based on the uniform delay assumption to obtain analytically tractable stability conditions, the proposed RL controller itself does not rely on this assumption. The controller learns a general stabilization

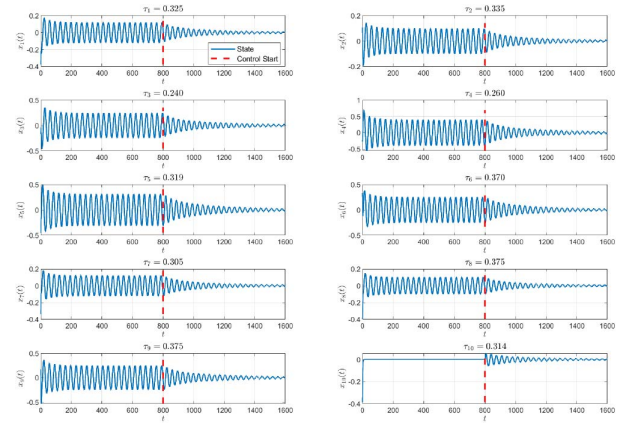


Fig. 22. Performance of the controller trained with $\tau = 0.3$ in a heterogeneous delay environment.

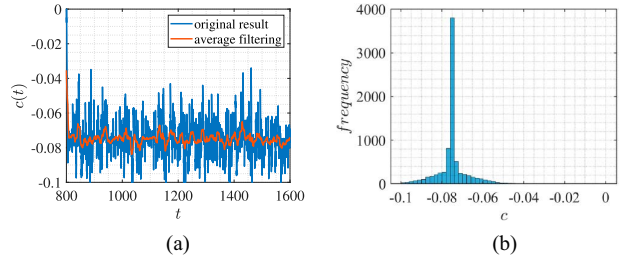


Fig. 23. Adaptive regulation and action distribution of the higher order coupling coefficient c by the RL controller when $\tau = 0.2$ for $t > 800$. (a) Temporal evolution of the coupling coefficient c , showcasing the agent's intervention starting at $t = 800$ to suppress system oscillations. (b) Distribution frequency of the control action c , illustrating the preference of the learned policy for specific coupling strength values.

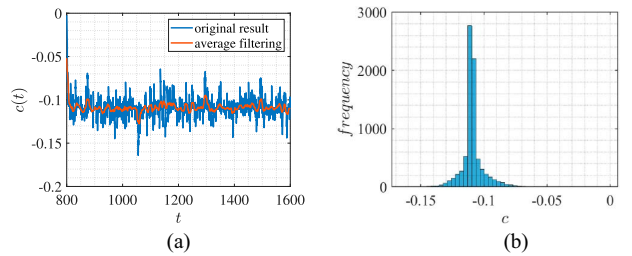


Fig. 24. Adaptive regulation and action distribution of the higher order coupling coefficient c by the RL controller when $\tau = 0.3$ for $t > 800$. (a) Temporal evolution of the coupling coefficient c , showcasing the agent's intervention starting at $t = 800$ to suppress system oscillations. (b) Distribution frequency of the control action c , illustrating the preference of the learned policy for specific coupling strength values.

strategy that adjusts the higher order coupling strength based on state feedback through interaction with the environment. Its robustness to delay heterogeneity stems from its model-free nature: it does not rely on an explicit model of the system dynamics or delay structure, but adapts to the actual dynamics by observing the system state in real time and outputting corresponding actions. Of course, while the delay variation range in this experiment ($\pm 30\%$) already covers the variation degree

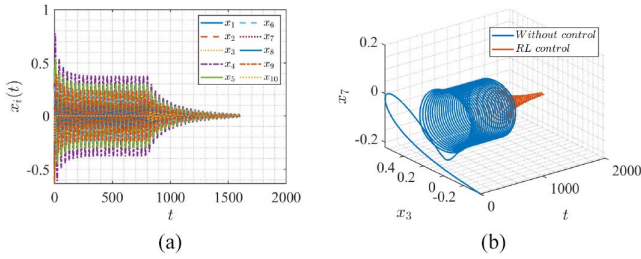


Fig. 25. Temporal waveforms and phase representations for system (1) comprising ten neurons under parameters $\mu = 0.58$ and $\tau = 0.3453$. Hopf bifurcation suppression via intelligent agent-controlled higher order coupling strength: $c = -0.1$ for $t < 800$ and RL-based regulation of c after $t = 800$. (a) Dynamic evolution of neuronal states x_i (where $i = 1, \dots, 10$) as functions of time t . (b) Time courses of specific neurons x_3 and x_7 .

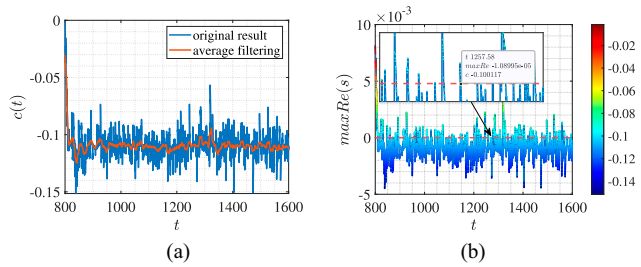


Fig. 26. Control action and spectral abscissa under the RL controller for the Hopf-threshold test in $\alpha = 1$, $\tau = \tau_0 = 0.3453$. The RL control is activated at $t = 800$. (a) Control trajectory $c(t)$. (b) Time evolution of the spectral abscissa $\max Re(s)$ computed from the characteristic roots using the instantaneous $c(t)$; the colorbar indicates the corresponding value of c . The dashed line denotes $\max Re(s) = 0$.

of most delays, the performance under extreme variations or rapidly time-varying scenarios still needs further exploration, which is also an important direction for our future work.

Furthermore, we fix as follows: $\mu = 0.58$, $\alpha = 1$, $c = -0.1$ and $\tau = \tau_0 = 0.3453$ (see Table I), and then apply the RL agent trained at $\tau = 0.3$ from $t = 800$. As shown in Fig. 25, once control is activated, the oscillatory waveform/phase portrait is progressively suppressed and the state trajectory approaches the equilibrium. Correspondingly, Fig. 26(a) shows that the controller drives $c(t)$ into a negative neighborhood, and Fig. 26(b) shows that the spectral abscissa becomes negative, indicating that the dominant characteristic roots shift to the left half-plane. Importantly, the evolution of $c(t)$ is consistent with the evolution of $\max Re(s)$: the magnitude of c required during stabilization is of the same order as that required to push $\max Re(s)$ below zero. Therefore, selecting negative c yields a smaller tracking error (and hence a less negative r_{te}), which dominates the overall reward given $r_{te} \gg r_{ap}$. In this sense, the symmetric penalty $r_{ap} = -\rho_{ap} \tanh^2(c)$ functions as a magnitude regularizer rather than inducing a suboptimal sign bias.

To systematically verify the robustness of the controller to initial oscillatory amplitudes, we conduct multiple sets of comparative experiments, testing the controller's performance under different initial amplitude conditions. The results are shown in Fig. 27. These experiments consistently demonstrate

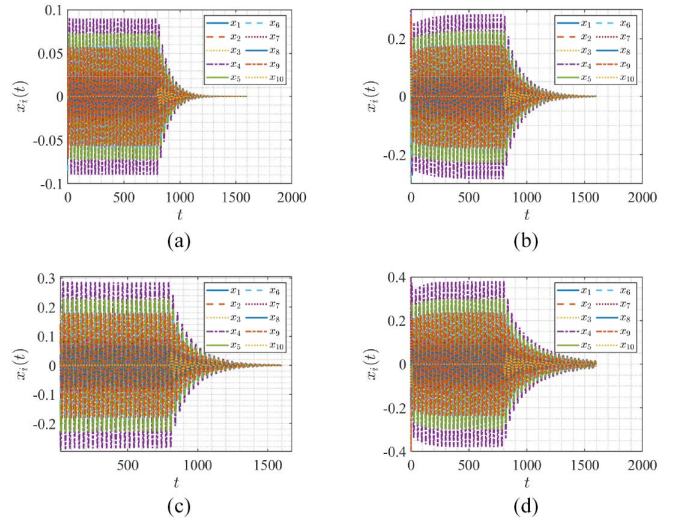


Fig. 27. State response $x(t)$ under the RL controller for four different initial conditions, producing four initial oscillation amplitudes. Control is activated at $t = 800$ in all cases: (a)–(d) correspond to increasing initial oscillation levels.

that the performance of the RL controller is highly robust to the initial oscillatory amplitude. The controller can dynamically adjust the higher order coupling coefficient c through online learning, shifting the system's characteristic roots to the left plane, thereby effectively suppressing oscillations and restoring stability.

VI. CONCLUSION

In this study, we have established an integrated theoretical and computational framework for analyzing and controlling the dynamics of large-scale modular higher order neural networks incorporating fractional-order derivatives and multiple time delays. We derived the characteristic equation of the high-dimensional system and established analytical criteria for local asymptotic stability and Hopf bifurcation. The bifurcation threshold is governed by a synergistic interplay of key system parameters, including the fractional order, self-feedback strength, higher order coupling strength, and network size.

We design a RL-based control strategy to adaptively regulate the higher order coupling coefficient, thereby enabling suppression of Hopf bifurcation. Through real-time interaction with the environment, the RL agent learned an effective policy for restoring system stability without dependence on parse expression. While the current method is effective for the network scales studied, the growth in feature dimension could become a bottleneck for extremely large networks. Future work will explore the following directions: feature selection and deep neural networks for automatic feature extraction to improve scalability; extensions to networks with heterogeneous and time-varying delays and the integration of deep RL architectures for improved generalization; and applications in neuromorphic computing and bio-inspired adaptive systems.

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